

Central composite design application in the optimization of the effect of pumice stone on lightweight concrete properties using RSM

Mujahid Ali^{a,b,*}, Abhinav Kumar^c, A. Yvaz^d, Bashir Salah^e

^a Department of Transport Systems, Traffic Engineering and Logistics, Faculty of Transport and Aviation Engineering, Silesian University of Technology, Krasińskiego 8 Street, 40-019 Katowice, Poland

^b Department of Civil Engineering, Faculty of Engineering, Universiti Malaya, Kuala Lumpur 50603, Malaysia

^c Department of Nuclear and Renewable Energy, Ural Federal University Named After the First President of Russia, Boris Yeltsin, 19 Mira Street, 620002 Ekaterinburg, Russia

^d World-Class Research Center "Advanced Digital Technologies", State Marine Technical University, Saint Petersburg 190121 Russia

^e Department of Industrial Engineering, College of Engineering, King Saud University, P.O. Box 800, Riyadh 11421, Saudi Arabia

ARTICLE INFO

Keywords:

Lightweight aggregate concrete
Compressive strength
Split tensile strength
Pumice stone
Response surface methodology
Central composite design application

ABSTRACT

Among other lightweight aggregates, one choice is pumice stone (PS), which can replace coarse aggregates in concrete. In this study, the natural coarse aggregates are replaced by various percentages of PS in concrete to examine its physical and mechanical properties. Experiments are designed to prepare mixtures (by varying ratios of PS) using response surface methodology (RSM). Slump and compaction factor tests are used to investigate the fresh properties, such as the workability of concrete. In contrast, the hardened concrete properties are determined using compressive strength (CS), flexural strength (FS), and split tensile strength (STS). The experimental, optimization and validation of the model results show that up to 30% of the PS can be replaced in the lightweight aggregate concrete for 7 – 56 curing days in which the CS is greater than 15 MPa, STS is 7–12% of the CS, and FS is 9 – 11% of the CS. Therefore, the optimum CS for the current study ranges from 17 to 20 MPa, in which the STS is 9.39 – 10.54% of the CS, and FS ranges from 17.49% to 21.12% of the CS. The quadratic model was proposed, and the ANOVA and model fitness was produced through RSM, with the coefficient of determination (R^2) above 0.99 (99%) for all three responses representing the high importance of the suggested model. Besides, the difference among the R^2 and adjusted R^2 was negligible, and the P-Value for CS, STS, and FS was below 0.05 (<0.0001), indicating the model's high relevance. RSM was used to improve the model, which was then verified in further lab trials.

1. Introduction

The use for construction materials demand is increased due to infrastructure development worldwide. Natural coarse and fine aggregates are the main constituents of construction and building materials. Due to its versatile concrete applications, rapid

* Corresponding author at: Department of Transport Systems, Traffic Engineering and Logistics, Faculty of Transport and Aviation Engineering, Silesian University of Technology, Krasińskiego 8 Street, 40-019 Katowice, Poland.

E-mail address: mujahid_19001704@utp.edu.my (M. Ali).

<https://doi.org/10.1016/j.cscm.2023.e01958>

Received 2 January 2023; Received in revised form 9 February 2023; Accepted 23 February 2023

Available online 24 February 2023

2214-5095/© 2023 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

urbanization, and industrialization, it is the material of choice for mankind. Portland Cement, coarse and fine aggregate are the three main forms of modern concrete that have a specific role in the concrete mix design and are mostly generated from natural resources. Due to rapid infrastructure and development, concrete is used in abundance, degrading natural resources. Therefore, the researchers are now looking forward to reducing the use of natural coarse aggregates by replacing them at varying percentages [1,2] or fully [3] in concrete with different other abundant nature aggregates such as pumice stone [4].

In recent and past studies, several industrial wastes are utilized in the concrete as a cement replacement, such as ash from timber [5], fly ash [6], glass polishing power [7], wood fiber waste [8], rice husk ash [9], limestone powder [10], demolition materials [11], calcium carbonate residues [12], and ceramic powder wastes [13]. However, some waste is used as a natural fine aggregate replacement, such as waste foundry sand [14], recycled glass [15], plastic [16], and rubber [17]. Besides, E-plastic waste [1], recycled concrete aggregate and ceramic waste [18], wood waste [19], and agro-waste [2] are used as a replacement for pure grainy aggregate in the concrete mix design and investigate the concrete properties. Regardless of the waste mentioned above, materials and industrial by-product waste, Lightweight Aggregate Pumice Stone (PS) are also highlighted in the literature as a replacement of pure coarse aggregate in concrete.

Several studies are conducted on PS worldwide as a natural coarse aggregate replacement to investigate the strength properties, economic, and environmental benefits. Regarding the economic benefits, its vary from country to country. For instance, M. Saltan and F. Selcan Findik (2004) conducted a study on the use of PS in flexible pavement and conclude the cost of 4.500\$ per unit length for Turkey [20]. Besides, other researchers concluded ranging between \$4 - \$10 whereas natural stones are usually between \$10 - \$15. Using PS as a preferential substitute for natural coarse aggregate can provide environmental as well as economic benefits by successfully utilizing waste resources that are available in abundant [21].

ACI 213R-87, 1998 defines lightweight structural concrete as concrete containing a density among 1350 and 1900 kg/m³ and minimum strength properties of 17 MPa. Over the last twenty years, there has been a significant increase in the understanding and implementation of high-strength, lightweight concrete based on artificial lightweight particles [22–24]. Pumice is a volcanic sponge-like substance formed by abruptly freezing magma and catching millions of microscopic air bubbles [25,26]. Pumice aggregates are widespread on the edges of volcanic regions, notably in the Mediterranean region, the majority of Indonesia and Turkey, and the Rocky Mountains of the United States [27]. However, modest research has proven that structural concrete with strength properties up to 25 MPa may be manufactured with sufficient economic advantages [28–30].

Idi et al., and Sultan et al., also concluded that pumice stones have produced a slightly lower compressive strength associated with standard concrete meaning that there is a demand intended for additional materials to maintain the compressive strength at a lighter weight [21,31]. However, up to 50% replacement, the concrete incorporating PS can achieve 25 MPa strength and can be used for structural purposes whereas, beyond 50% replacement can be used for non-structural purposes. When lightweight construction concrete with a pumice stone is employed to build and maintain civil structural components, the advantages of lower total self-weight, fire resistance can be attained, decreased natural resource depletion (economic advantages), and improved sound insulation and heat [23]. Some researchers used the pumice stone as a fine aggregate replacement in mortar to evaluate the shrinkage (micro-cracks) in a mortar [32–35]. Liu et al. investigate the effect of pumices with varying water absorption rates used to substitute river sand by 10–30% on the qualities of the created Ultra-High Performance Concrete (UHPC). They discovered that using pre-wetting pumice (particle size distribution of 0.6–1.25 mm) may greatly minimize autogenous shrinkage of UHPC and improve the interfacial transition zone (ITZ) skeleton among UHPC paste and hydrous pumice while maintaining engineering properties [36].

Karthika et al. used pumice stone in concrete as a coarse aggregate at a varying replacement ratio to evaluate the strength performance. They concluded that by using 50% of pumice lightweight aggregate concrete (PLWAC), the optimum CS, STS, and FS can be achieved and reduces the self-weight and absorbs more water due to the pores, which reduce the strength properties that can be used for non-load bearing walls, partition walls, and earthquake resistance structures [37]. Besides, Zhang and Gjorv focus on the strength controlling factor of high-strength lightweight concrete (HSLWC) to investigate the mechanical properties by using five different types of lightweight aggregates. They concluded that the STS/CS ratio is lower for HSLWC than HSNWC. The elastic modulus varies from 17.8 to 25.9 MPa and is much lower than HSNWC [38]. T. Parhizkar et al. investigated the structural cost, dead load reduction, foundation size, and reinforcement by using a pumice stone to partially replace coarse aggregate [39]. B. Devi Pravallika and K. Venkateswara Rao recommended improving the durability and strength attributes of M40 concrete by substituting coarse aggregate (0%, 10%, 20%, 30%, 40%, and 50%) with a natural lightweight aggregate pumice stone [40]. C. Krishnaveni and K. Guru Kesav Kumar investigated the use of pumice stone as a partial aggregate substitute in cement concrete [41]. Several studies have been conducted on the waste replacement, by-product, and replacement of natural resources over other abundantly available natural resources to evaluate fresh, mechanical, and microstructural properties [42–48].

Response surface modeling (RSM) is a collection of computational and scientific methods that are used to examine the interplay of several exogenous variables and the response variable (s) of a physical system. Central Composite Design (CCD) is used in the design expert software to design and optimize the experiments. After getting the mix design from the RSM – CCD, the designed experiments are performed. The experimental values are used as input parameters for the linear regression to build the statistical model. The curing days and percentage replacement are used as input parameters, and the strength properties such as CS, STS, and FS were set as response variables. Several researchers used RSM for their experimental design and optimization in recent studies. For instance, Zahid et al. used RSM to predict the engineering properties of geopolymer composite by replacing it with Ordinary Portland Cement (OPC). Ali et al. used the design and statistical modeling of concrete incorporating Waste Foundry Sand (WFS) as a sand replacement to evaluate the strength properties [49]. Khan et al. used the partial substitution of OPC by silica fume and irradiated waste polyethylene terephthalate in various quantities to evaluate the impact on semi-flexible payment [16,50]. Moreover, Al Salaheen et al. used response surface methods to estimate and optimize mortar compressive strength by using heat-treated fly oil shale ash as an effective supplemental

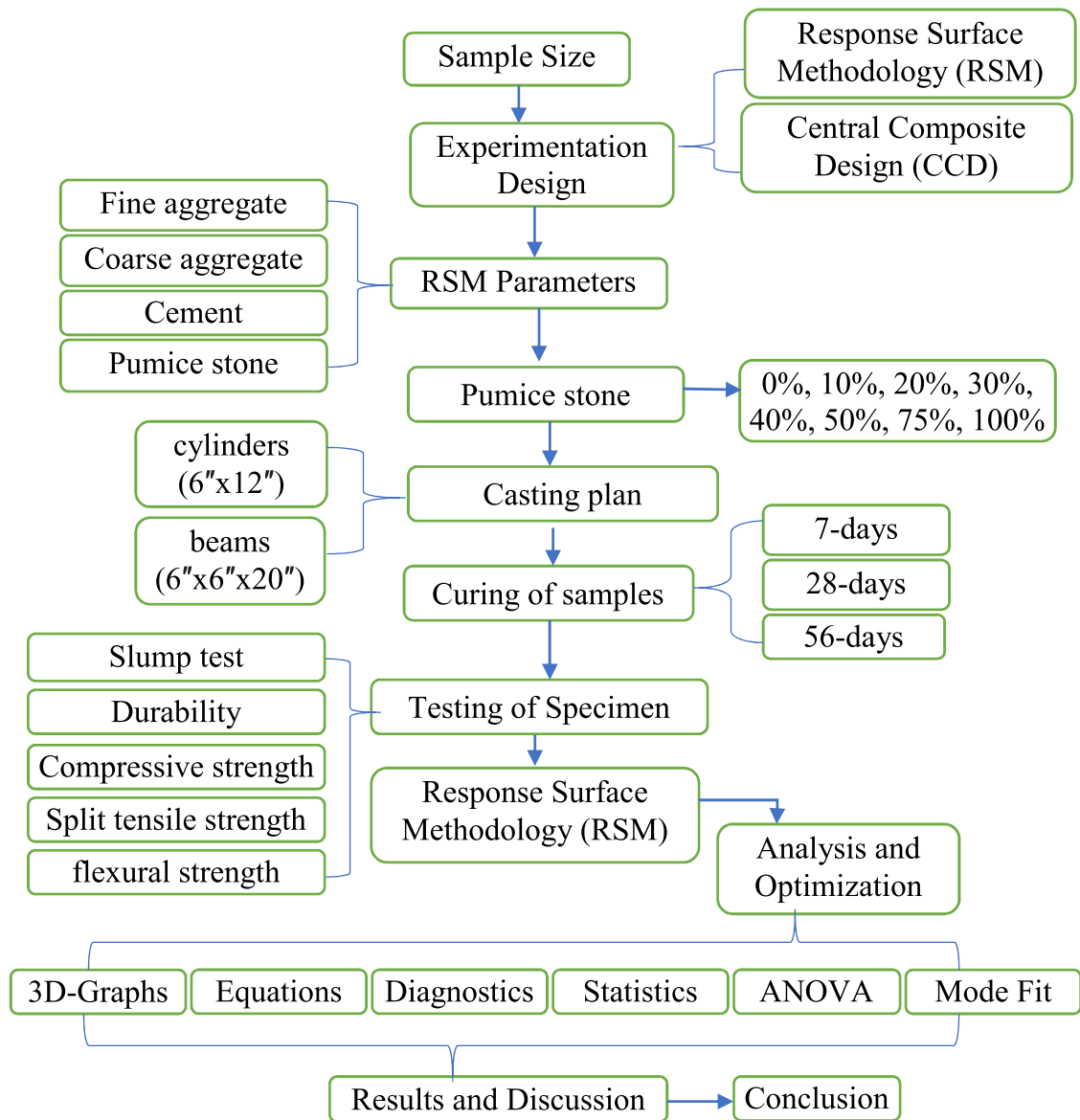


Fig. 1. Methodology Flow Chart.

cementitious material [51]. Fediuk and Ali summarized several articles in their editorial article that used RSM for statistical analysis, experimental design, and optimization [52]. Memon et al. investigate the morphological, physicochemical, and rheological properties of bitumen-containing petroleum sludge using RSM [53].

Based on the recent and relevant literature, it is vital to examine the influence of natural coarse aggregate as a partial and full replacement in concrete with PS to evaluate the mechanical and physical properties incorporating lightweight aggregate PS. Despite the availability of research papers on the replacement of lightweight aggregate incorporating PS in ordinary concrete mixes, optimization, and statistical modeling of the concrete mixture have not been carried out. As a result, the current work attempts to fill such gaps in the existing literature by investigating the physical and mechanical characteristics of lightweight aggregate PS as a coarse aggregate substitute in concrete using the experimental design matrix provided by RSM's CCD feature. Besides, statistical tools (RSM) are designed to optimize the experimental and concrete mix design. Lastly, using pre-defined design criteria, multi-objective design optimization was used to determine the of PS and number of curing days for optimum slump and mechanical characteristics. Additional experiments were conducted to validate the optimal results.

2. Methods and materials

This section contains experimental information used to examine the physical and mechanical characteristics of lightweight

Table 1
Physical and Chemical Composition of OPC.

Chemical composition of OPC			Physical Properties of OPC		
Constituent	Value (%)	Standards (%)	Constituents	Observed	Typical
CaO	62.90	60–68%	SG	3.09	3–3.15
SiO ₂	23.10	16–24%	SS	322 m ² /kg	329 m ² /kg
Al ₂ O ₃	4.90	3–8%	Consistency	25.9%	21–20%
Fe ₂ O ₃	3.30	0.49–6.01%	Initial Timing	55 min	> 30
SO ₃	1.80	0.99–3.01%	Final Timing	230 min	< 600
MgO	1.79	0.12–3.99%			
K ₂ O	1.05	0.50–1.01%			
Na ₂ O	0.31	< 1%			
LOI	0.66	< 1%			

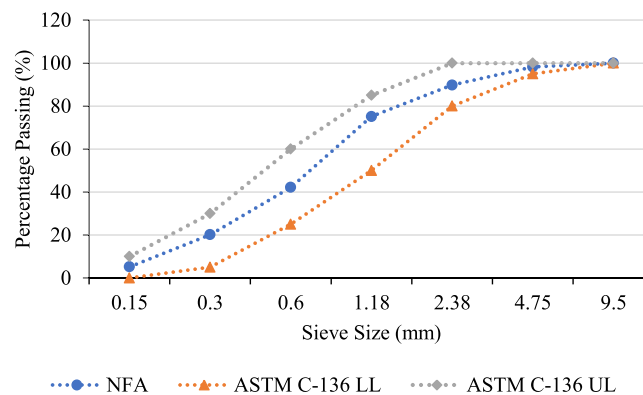


Fig. 2. Percentage passing of NFA with LL and UL as per ASTM C-33.

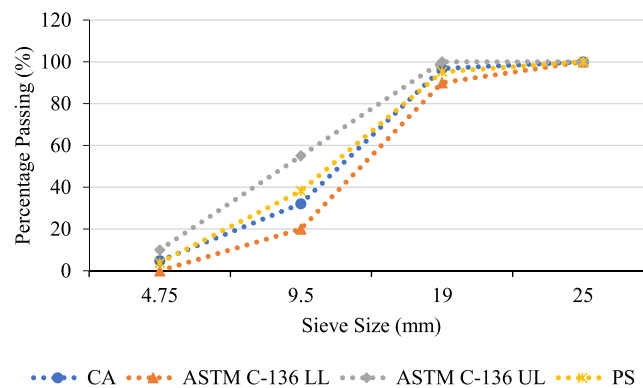


Fig. 3. PS and CA with LL and UL of ASTM C-33.

concrete including pumice stone. The total amount and composition of fine aggregate, coarse aggregate, and pumice aggregates as a coarse aggregate replacement from 0% to 100% (0%, 10%, 20%, 30%, 50%, 75%, 100%), and cementitious materials, as well as air contents, are material variables. These materials are described in detail. The flowchart shown in Fig. 1 illustrates the research methodology flowchart.

This study employed Ordinary Portland Cement of Type-1 (OPC, Type-1) in accordance with the American Society of Testing and Materials (ASTM) Part C-150 ASTM-C-150 standard [54] in the concrete mix design. In addition, before using it in the concrete mix design, it is vital to investigate the basic properties of the material. Therefore, the cement consistency and specific gravity (SG) are evaluated as per ASTM C-188 and recorded as 25.9% and 3.09. As per the standards, the standard consistency of cement ranges 25-and 35%, whereas the SG is 3–3.16; however, both the consistency and the SG are within the ASTM standards' limits. Furthermore, the Vicat needle apparatus is used, as per ASTM C191–19 [55], to examine OPC's final and initial setting time. As a result, OPC's final and initial setting time is achieved at 230 min and 55 min. In contrast, the standard limits are 30 min (minimum) and 600 min (maximum) for the OPC to achieve its initial and final setting time. The physical and chemical composition of OPC is shown in Table 1. The most



Fig. 4. Pumice stone.

Table 2

Chemical and physical properties of PS.

Physical composition		Chemical composition	
Constituents	Value	Constituents	wt%
Slurry pH	6.0	SiO ₂	68.3
pH	5.0	Al ₂ O ₃	14.59
r _p (A)	2.0	Fe ₂ O ₃	2.58
Moisture (%)	1.0	Na ₂ O	4.890
Bulk density (g/cm ³)	0.85	K ₂ O	3.62
V _p (cm ³ /g)	0.09	CaO	1.96
S _{BET} (m ² /g)	32	MgO	1.09
		MnO	0.219
		TiO ₂	0.14
		LOI	2.61

contributing composition is CaO and SiO₂ in the OPC chemical composition, which is 86% in total. The current research on concrete mix design employs locally accessible fine aggregate. Therefore, the particle size distribution or fineness modulus of natural fine aggregate (NFA) is estimated via sieve analysis. As per the ASTM C-136–06/C-33 M [56], the lower limit (LL) and the upper limit (UL) of the standard are compared with the percentage passing of NFA, as shown in Fig. 2. The fineness modulus of the NFA was found to be within the ASTM UL and LL, indicating high NFA quality; nevertheless, in some cases, it was close to UL, while in others, it was found to be contrary.

In the current investigation, a coarse aggregate of locally available crushed angular maximum size retained 3/8 size is chosen as a coarse aggregate in the concrete mix design. Sieve analysis for coarse aggregates is performed in the laboratory using an electric sieve shaker, according to ASTM C136–14 [57]. Fig. 3 depicts the ASTM gradation curves and the ASTM UL and LL of the ASTM.

Pumice is a light-colored porous igneous rock created through explosive volcanic eruptions when molten lava containing many gas bubbles is hurled into the air. When it cools, it hardens into pumice. Pumice stone (PS) has numerous physical properties. It became an extraordinarily light rock due to the many permeable bubbles/holes inside it. With an average porosity of 90%, PS are lightweight and initially float in water [58]. The density of pumice varies according to the thickness of the solid substance between the bubbles. It's used in various industrial and consumer products as a lightweight concrete aggregate, landscaping material, and abrasive [59]. The PS was gathered from one of the US apparel industries that are located in Lahore city of Pakistan as shown in Fig. 4.

The chemical and physical properties of PS are depicted in Table 2. The chemical composition of pumice varies depending on the deposit, but it is mostly composed of aluminum oxide, silicon dioxide (amorphous aluminum silicate), and residue volumes of additional oxides. The pumice stone has the largest SiO₂ content compared to the other constituents, and it is a mixture of Si, Al, Ca, Mg, and Fe oxides [60]. The primary constituents, such as SiO₂, Al₂O₃, and Na₂O, constitute 87.78% of the total, while the remaining constituents provide 12.22%.

To assess the physical and mechanical qualities of concrete using a PS, its partially and completely replaced coarse aggregate at varying percentages in the concrete mix design. For the experimental design, approximately 350 kg of PS were gathered from the US apparel business in Lahore, Pakistan. Before using PS in the concrete mix design, the PSD of PS have been evaluated using an electronic sieve shaker. Fig. 3 depicts the sieve analysis of PS, coarse aggregate, and ASTM UL and LL.

According to ASTM standards, the CS, STS, and FS are evaluated after 7 days, 28 days, and 56 days of curing using a 0.45 water-cement ratio and 1:1.5:3 ratio. Therefore, the specimens are cast and kept for 7–56 days in the water tank to test the characteristics on 7–56 days along with varying percentages of PS as a coarse aggregate substitute.

2.1. Experimental design based on RSM

RSM is a powerful statistical method to design, analyze, optimize, and validate experimental results. Most often, it is used in the concrete industry for the experimental design with fluctuating proportions of waste replacement as a natural resource to find out the association among the manipulated variable and response variables. For the experimental design of the current study, Central

Table 3
Input Ranges Parameters for the CCD.

Input Variables	Lower Limit	Upper Limit
Pumice Stone (% ratio)	0%	100%
Curing days	7 days	56 days



Fig. 5. (a) casting and coding of the cylinder, (b) curing of a cylinder containing PS.

Composite Design (CCD) technique tool was used in the Design Expert software version 11.1.0 as it provides outstanding competencies for designing an idyllic experiment on a mixture or combination of factors, components, and processes. Primarily, the researchers provide the input and the output variables to the design expert RSM to generate the experimental design, so in the current study, the percentage replacement of PS as a coarse aggregate and curing days (7, 28, and 56 days) were used as input variables and the response variables was the strength properties of concrete such as CS, STS, and FS. Ali et al., 2022 use the CCD tool of RSM in Design Expert software to investigate the strength properties of concrete incorporating waste foundry sand as a fine aggregate replacement [49]. The top and lower ranges for such input parameters were set at 0–100 for % PS replacement and 7–56 days for curing. Table 3 depicts the range parameters that are used in CCD for the statistical analysis and experimental design.

Based on model relevance, quadratic polynomials, and linear models were offered to anticipate the mechanical characteristics of concrete. To forecast the responses, a 2nd-degree polynomial equation was utilized, and the generalized equation for the anticipated model used is provided in Eq. 1.

$$Y = C + A_1(X_1) + A_2(X_2) + A_3(X_1^2) + A_4(X_2^2) + A_1(X_1 * X_2) \quad (1)$$

Whereas Y represents the response variables that are CS, STS, and FS. Besides, X_1 and X_2 show the manipulated variable, for example, curing days and percentage replacement of PS, whereas C is the constant intercept, and A_1 and A_2 are the equation's coefficients.

2.2. Concrete mix design

The PS was substituted as a coarse aggregate with varying percentages of 0%, 10%, 20%, 30%, 50%, 75%, and 100%, producing a total of seven different mixtures. Depending on the trials, a sum of 189 specimens being produced and prepared in twenty-one (21) cylinders for CS on seven days, with three samples created for 0%, 10%, 20%, 30%, 50%, 75%, and 100% (to take the average value for the specific specimen to avoid human and experimental errors) which makes a total of twenty-one cylinders. The same procedure was followed for 28 and 56 days of curing. To test the FS and STS of lightweight concrete using PS, twenty-one cylinders were produced and arranged for the same proportion and amount of curing days. The cylinder mold size utilized for concrete casting was 6" x 12" for the CS and STS, and 150 × 150 × 500 mm for the FS. Fig. 5(a) depicts the casting and coding of the cylinder, whereas Fig. 5(b) shows the curing of a cylinder containing PS.

2.3. Compressive strength (CS)

CS tests on concrete specimens should be performed using a universal testing machine (UTM) to evaluate the load-carrying capability under compression. The bottom and top sides of the column specimen are sealed with plaster of Paris with a depth of 12 millimeters to adequately disperse the stress. A steel plate with a 9"x9" cross-sectional area is also put just on the upper portion of the column sample to ensure consistent load distribution. The specimen is positioned among the jars such that the filled face approaches the testing equipment. Vertical loads are applied at a constant rate till failure happens, and the highest load reading at the failure stage is reported. The CS test is performed using the ASTM C39 standard test procedure [61].

2.4. Split tensile strength (STS)

As mentioned earlier, a total of 63 samples were cast for the STS at varying percentages ranging from 0% to 100% and curing days ranging from 7 to 56 days. The cylinder dimension was the same as used for the CS. The STS is mostly calculated and recorded as 7 –

Table 4
Slump values of lightweight aggregate concrete incorporating PS.

Percentage Replacement	Slump Value (mm)
Control mix/Normal concrete (0%)	132
PS – 10%	102
PS – 20%	95
PS – 30%	79
PS – 50%	55
PS – 75%	40
PS – 100%	28

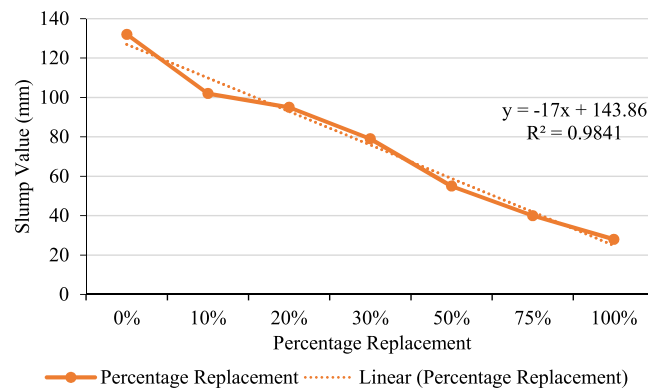


Fig. 6. Slump results at varying percentages of PS.

15% of the CS, even though it's an essential concrete parameter. Based on recent and past studies, the normal tensile strength of concrete is recorded ranging from 300 to 700 PSI (2 – 5 MPa), meaning that it is roughly 10% of the CS of concrete. Besides, the standard test method of ASTM C496 is used to perform the STS [62].

2.5. Flexural strength (FS)

Although several specimens and the curing days for the FS of concrete are similar to STS and CS, the beam's measurements differed. The prism beam was used to produce the samples for the FS with a dimension of 150 × 150 × 500 mm. As per modern and previous research, the normal FS for conventional concrete ranges from 3 to 5 MPa. The FS is carried out with the standard test method of ASTM C293 [63].

2.6. Water absorption

The water absorption is determined based on ASTM C 642–97. Water absorption of natural concrete specimens is lower compared to that of lightweight aggregate concrete (PS) samples. The high moisture content of lightweight aggregates at SSD circumstances causes a greater water/cement proportion in the transition zone (among paste and aggregates), resulting in enhanced permeability of the concretes.

2.7. Fresh concrete workability

The properly/improperly mixing batch can be accessed through the consistency and workability of concrete. The slump test is used to determine the consistency of fresh concrete components before they harden. Slump value and compaction factors are events that may be used to assess the workability of fresh concrete and, as a result, the speed with which concrete moves.

3. Results and discussion

This segment presents the experimental results obtained in the laboratory and through analytical analysis of RSM for the STS, FS, and CS. The drying days and percentage replacement were used as input variables in the RSM to predict CS, STS, and FS, in which each response variable is independently investigated. The optimization process of the RSM investigates the association among the drying days and PS percentage on the concrete strength properties, which contains multiple regression analyses. Several plots, for instance, actual vs. predicted, box-cox plots, normal plots of residuals, counter graphs, and 3D graphs were generated through RSM for each of the response variables. Besides, the regression analysis table, ANOVA, and equations are developed for each of the response variables

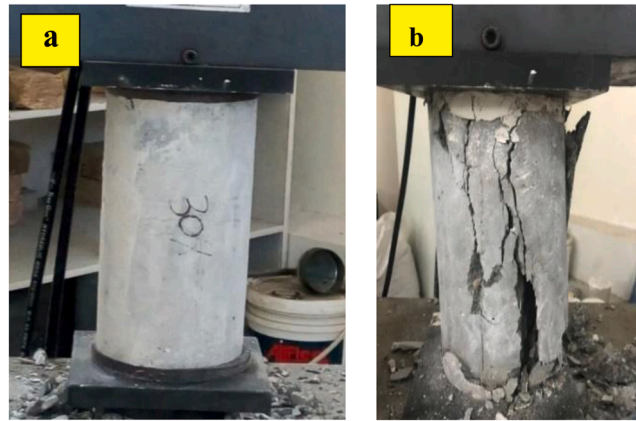


Fig. 7. (a) loading in UTM (b) crushing cylinders during the CS containing PS.

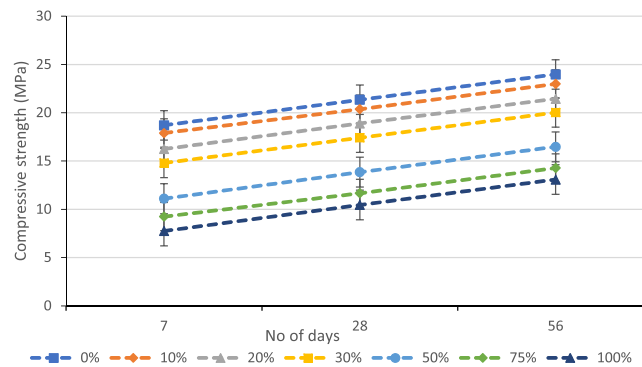


Fig. 8. Experimental results of CS containing PS at varying percentages and curing days.

and are discussed below.

3.1. Slump cone test

A slump cone test determines the fresh properties of lightweight aggregate concrete incorporating PS. The slump cone test is a popular method for determining the consistency of concrete. The experimental results show that the slump values decrease by increasing the percentage of the PS in lightweight aggregate concrete as natural coarse aggregate. The decrease in the slump values is maybe due to the regular shape and size of the PS. The wetter the mix, the higher will be the slump values [64]. Due to the proper shape of PS, the air voids or air entrained are properly filled up which causes in reduction of slump values. The highest slump value is found on the 10% replacement of the PS. Table 4 and Fig. 6 depict the slump values obtained at varying replacement percentages. The coefficient of determination (R^2) is 0.984 showing a 98.4% correlation between the slump values and percentage replacement.

3.2. Water absorption

The water absorption assessment is used to determine how much water the concrete absorbs. This test is performed to determine the permeability of concrete. The absorption rate of the PS-containing concrete is greater than that of the control mix with 0% PS substitution as a natural coarse aggregate. However, there is a variation in the absorption of the concrete containing PS at varying percentages due to their internal properties. The highest water absorption was found at 50% replacement of the PS in concrete. Karthika et al. concluded that the highest water absorption was found at 50% PLWAC, and beyond that, it was decreased [37], whereas Kale et al. investigated that the increase of PS in concrete caused a rise in water absorption of concrete due to its pores [65]. The fall and rise in the water absorption of concrete incorporating PS are due to the microstructure properties of the PS. The aggregate used in the current study is in wet conditions with surface moisture of 3.5%. Furthermore, the moisture level of the aggregate must be assessed earlier to mixing because it influences the quantity of mixed water necessary to attain the requisite water/cementitious materials ratio [66].

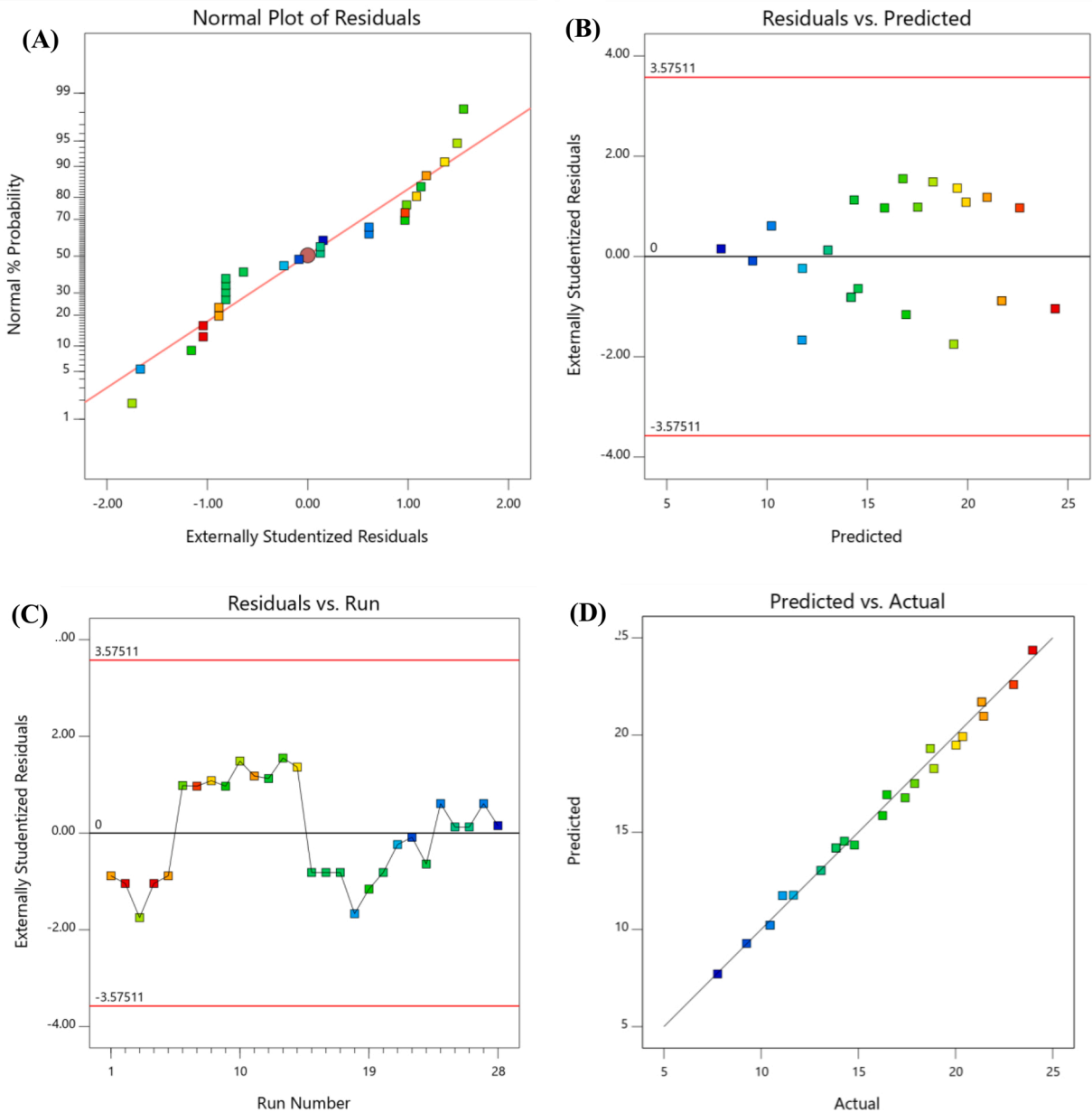


Fig. 9. Distribution of Data for CS; (A) Normal Plot of Residuals; (B) Predicted vs. Residuals; (C) Run vs. Predicted; (D) Residuals vs. Actual.

3.3. Compressive strength (CS)

As discussed earlier, several samples at a varying percentage of PS as a full and partial substitute of natural coarse aggregate are prepared to evaluate the CS of concrete incorporating lightweight aggregate PS on 7–56 days of drying. Fig. 7(a) depicts the CS of concrete samples while loading, where Fig. 7(b) shows the crushing cylinders during the CS containing PS at varying percentages and curing days. The experimental results of the CS are shown in Fig. 8. After the analytical results that are generated through RSM are compared with the experimental results, extra experiments were conducted to optimize and validate the model for the CS with specific PS ratio and curing days.

The experimental results show that by increasing the ratio of the PS in the concrete as a natural coarse aggregate replacement, the CS steadily decreases. However, up to 30% replacement, the CS for 7–56 days is within the ranges of lightweight weight aggregate strength which is 15 MPa. Lightweight structural concrete, according to Wu et al., is described as concrete having an oven-dry density among 1600 and 2000 kg/m³ and a CS greater than 15 MPa [67]. As a result, this study meets the standards for lightweight structural concrete, which is congruent with the definition published by CEB/RILEM [68]. On the other side, the control mix concrete is higher

Table 5
ANOVA and fit statistics.

Responses	Compressive strength (MPa)	Split tensile strength (MPa)	Flexural strength (MPa)
Standard Deviation	0.4482	0.0443	0.0627
Mean	16.16	1.62	3.36
R ²	0.9923	0.9913	0.9935
Adjusted R ²	0.9906	0.9893	0.9920
Predicted R ²	0.9882	0.9864	0.9891
Adequate Precision	80.2801	70.9810	89.1417
Model F-value	569.69	499.71	670.53
Model p-value	< 0.0001	< 0.0001	< 0.0001
Proposed/Suggested Model	Quadratic	Quadratic	Quadratic
Model Remarks	Significant	Significant	Significant

Table 6
Coefficients of the predicted equation.

Coefficients of Equation		Response		
		Compressive strength (MPa)	Split tensile strength (MPa)	Flexural strength (MPa)
Constant	C	+ 18.4345	+ 1.9452	+ 2.9147
PS	A1	-0.1871	-0.0195	-0.0197
CA	A2	+ 0.1276	+ 0.0128	+ 0.0381
PS x CA	A3	+ 0.00005	+ 0.00001	-0.000005
PS ²	A4	+ 0.0007	+ 0.00008	+ 0.00008
CA ²	A5	-0.0004	-0.0001	-0.00013

than the lightweight aggregate incorporating PS as a full and partial replacement for all 7–56 days.

The CS on 7 days of curing with 10%, 20%, and 30% replacement are 17.9 MPa, 16.25 MPa, and 14.8 MPa, respectively, which falls in the definition of lightweight aggregate concrete. On the other side, Hedjazi concluded that to be considered lightweight structural concrete (SLWC), the minimum 28-day compressive strength is 17 MPa [69]. Therefore, the concrete CS for 28 days of curing and 10%, 20%, and 30% replacement of PS are higher than 17 MPa, which can be considered SLWC. Besides, the CS of control mix concrete on 28 days must be greater than 17 MPa, which in the current study is 21.35 MPa. Therefore, the current study concluded that up to 30% of the PS can be replaced by natural coarse aggregate in the lightweight concrete aggregate for 7–56 days of drying and can be used as SLWC. Beyond 30% replacement, it is suitable to be used for the partition walls and non-load bearing walls, which reduce the overall self-weight of the structure. It is well-known that the CS increases by increasing the curing days, and optimum strength (95%) can be achieved on 28 days of curing. The current study proves that the CS on 56 days was higher than the rest of the curing days for all varying percentages.

The current study evaluated the correlation between the multiple inputs and output parameters using RSM. Curing days and percentage replacement were treated as input variables, whereas strength properties such as STS, FS, and CS were the output parameters in RSM. Several plots, for instance, residuals vs. predicted, residuals vs. the run, predicated vs. actual, 3D graphs, normal plots of residuals, and contour graphs.

Fig. 9 (A) depicts the normal plots of residuals where (B) Predicted vs. Residuals; (C) Run vs. Predicted; and (D) Residuals vs. Actual, which shows the data are distributed normally through the straight line between the points. The closer the point to the line shows the normal distribution of the data, whereas far is the opposite. As a consequence of the results, it is clear that the model is adequate for the purpose and be able to be employed to determine the optimal extraction parameters. The model's projected values are generated in design expert software using the prediction equation, while the real values are achieved through experiment runs.

Table 5 shows the ANOVA and statistics of fit analysis for the STS, FS, and CS. Several terms such as standard deviation, mean, coefficient of determination (R²), Predicted R², Model F, and P-values, Adjusted R² are generated through RSM for the STS, FS, and CS. R² and P-value of the model is the most important factor in accessing the model. For the current study, the R² values for the CS are noticed as 0.9923, suggesting that it defines 99.23% of the change and is statistically substantial. However, it is 0.9913 (99.13%) for the STS and 0.9935 (99.35%) for the FS. Recent and past studies concluded that an R² of 0.80 and above is acceptable for the correlation between the variables [70–72]. The model's significance can also be seen from the variation among the adjusted R² and R². The smaller the difference between the values reveals the superior significance of the model, which is very minimal for the current study for all the response variables.

The P-value should be less than 0.05 to claim the significance of the model. However, the current study P-value for all the response variables is less than 0.05, in which CS, STS, and FS show less than even 0.0001, indicating the superior significance of the model. Furthermore, the model is tested for significance in terms of R², Adjusted R², and P-values, which show significance in every aspect; therefore, in the remarks, the model is significant.

The current study also develops coded equations for the STS, FS, and CS from RSM. The generalized quadratic polynomial Eq. 2 predicts the response value (i.e., Dynamic Modulus).

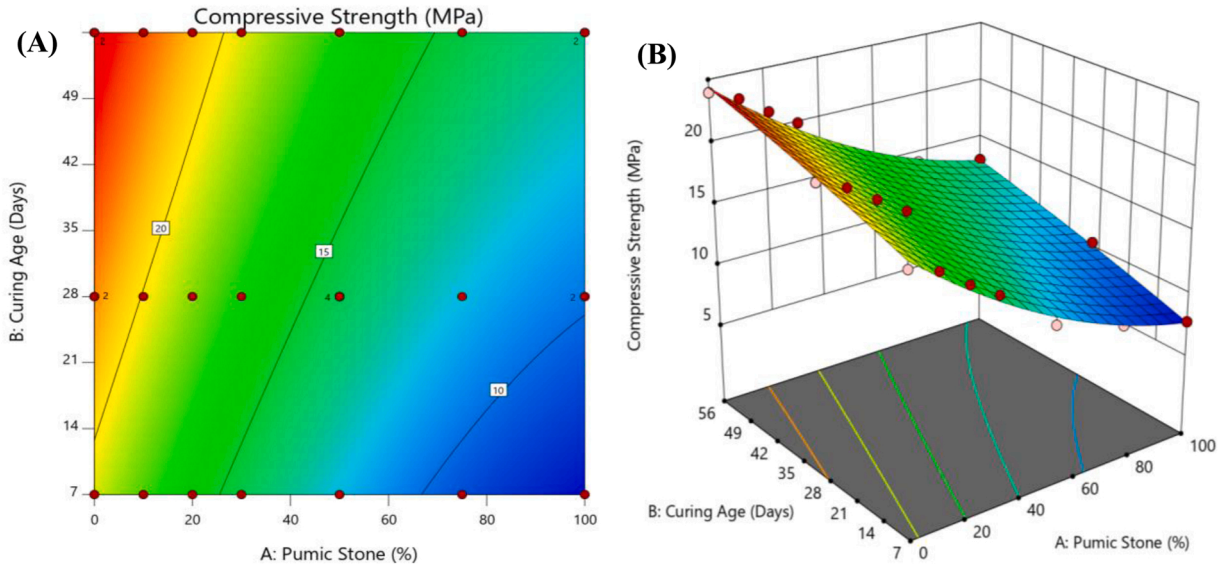


Fig. 10. (A) Contour and (B) 3D Graph for the CS of concrete incorporating PS.

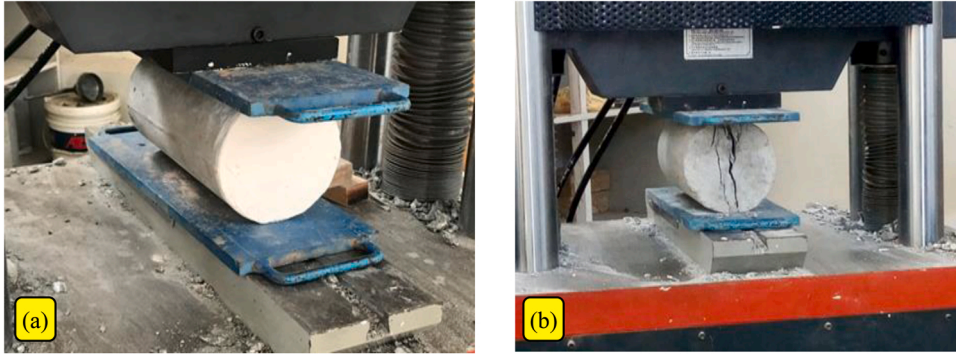


Fig. 11. (a) placement of concrete cylinder in UTM (b) crushed cylinder during the STS.

$$Y = C + A_1(X_1) + A_2(X_2) + A_3(X_1 * X_2) + A_4(X_1^2) + A_5(X_2^2) \quad (2)$$

Where Y= Predicted responses (CS, STS, and FS), X1 and X2 = Independent variables pumice stone (PS) and curing age (CA)), C= interception constant, A1 and A2 = Linear coefficients, A3 = Interaction coefficient, and A4 and A5 = Quadratic coefficients.

The above equation can be rewritten as Eq. 3; while considering the dependent and independent variables used in the current study.

$$CS/STS/FS = C + A1(PS) + A2(CA) + A3(PS * CA) + A4(PS^2) + A5(CA^2) \quad (3)$$

Where STS, FS, and CS are used for split tensile, flexural strength, and compressive, PS for pumice stone, and CA for curing age. Table 6 depicts the coefficients of equations corresponding to the response variables.

After including the values of the constant, A1 – A5, the final equation for the CS, STS, and FS is developed. Eq. 4 shows the compressive strength, whereas Eqs. 5 and 6 shows the split tensile and flexural strength generated from RSM using the quadratic model. The negative and positive term signs of the equation indicate the antagonistic and synergistic influence of external factors on the outcome in statistical analysis. From Eqs. 4 – 6, it can be seen that the PS synergistically affect the STS, FS, and CS of lightweight aggregate concrete.

$$CS = +18.4345 - 0.1871(PS) + 0.1276(CA) + 0.00005(PS * CA) + 0.0007(PS^2) - 0.0004(CA^2) \quad (4)$$

$$STS = +1.9452 - 0.0195(PS) + 0.0128(CA) + 0.00001(PS * CA) + 0.00008(PS^2) - 0.0001(CA^2) \quad (5)$$

$$FS = +2.9147 - 0.0197(PS) + 0.0381(CA) + 0.000005(PS * CA) + 0.00008(PS^2) - 0.00013(CA^2) \quad (6)$$

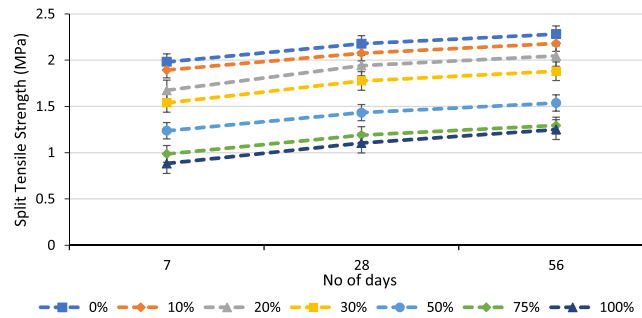


Fig. 12. Experimental results of STS containing PS at varying percentages and curing days.

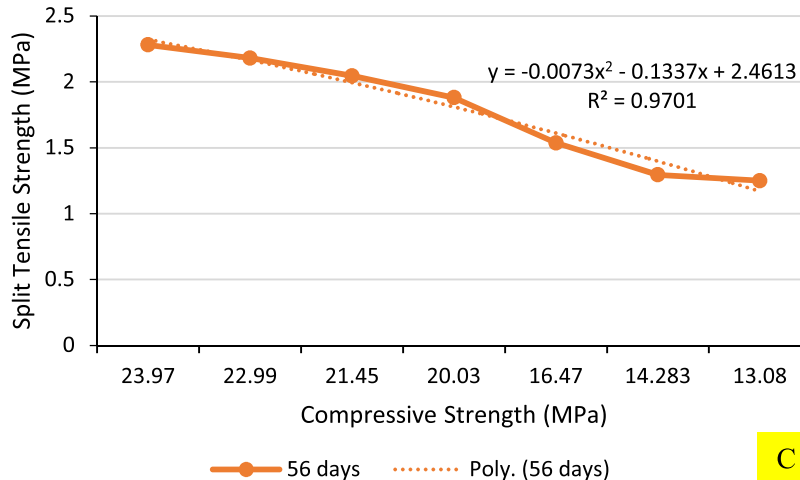
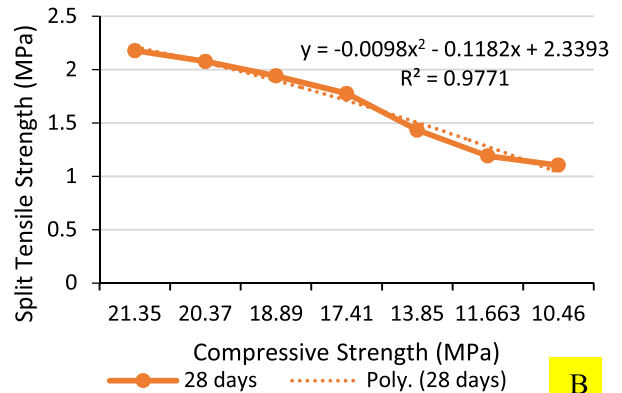
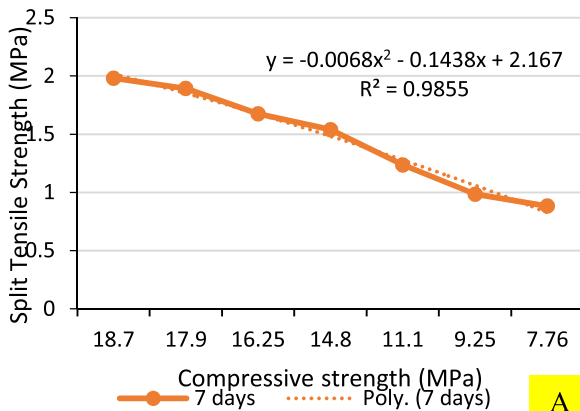


Fig. 13. Correlation between CS and STS a) 7-days b) 28-days c) 56-days.

In the design expert software, the interaction between the CS, curing ages, and PS as a coarse aggregate replacement is shown in the form of contour and 3D graphs, as indicated in Fig. 10 (A) and (B). In the 3D graph, the curing days are represented in the x-axis, the CS is denoted in the y-direction, whereas the replacement of PS is represented in the z-axis; however, it was the opposite in the contour graphs as shown in Fig. 10 (A). It can be seen from the graphs that the CS is increasing by increasing the curing age but decreasing with an increment of the ratio of PS as a coarse aggregate replacement. For example, the maximum CS was obtained on 56 curing days with 0% replacement which is 23.97 MPa; however, up to 30% replacement and all curing days, the CS of lightweight weight aggregate concrete incorporating PS were within the standard limits that in 17 MPa.

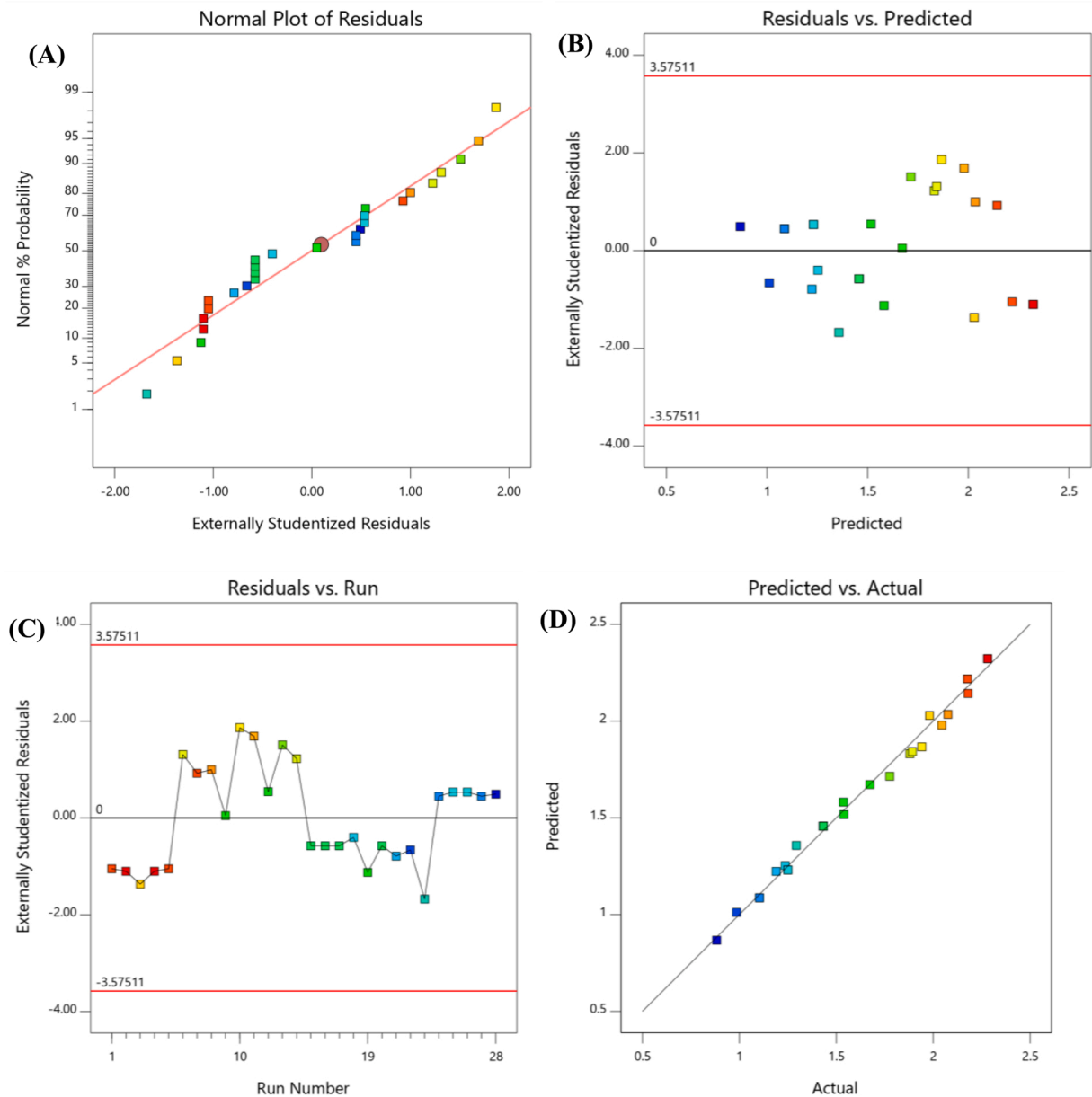


Fig. 14. Distribution of Data for STS; (A) Normal Plot of Residuals; (B) Predicted vs. Residuals; (C) Run vs. Predicted; (D) Actual vs. Residuals.

3.4. Split tensile strength (STS)

The STS of concrete including PS as a coarse aggregate substitute are experimentally assessed at different percentages and curing days. RSM was applied to the experimental data to find the correlation between the variables and compare the analytical results with the experimental results. Fig. 11 (a) shows the placement of the concrete cylinder in UTM, whereas (b) illustrates the crushed cylinder during the STS. Moreover, Fig. 12 depicts the experimental results of STS of concrete containing PS at varying percentages and drying days. Although STS is a main strength property of concrete, it depicts just 7–15% of its CS in the case of conventional concrete. By increasing the lightweight aggregate PS ratio, the experimental findings reveal a progressive reduction in the STS of concrete during all 7 days, 28 days, and 56 days of curing. The control mix design for all the curing days is higher than the percentages substitute of PS as natural coarse aggregate replacement.

The STS on 7 days of curing with a 10%, 20%, 30%, and 50% replacement were recorded as 1.893 MPa, 1.674 MPa, 1.539 MPa, and 1.237 MPa that were 10.57%, 10.30%, 10.39%, and 11.14% of the CS of LWAC incorporating PS. However, on the 28 curing days, 2.076 MPa, 1.941 MPa, 1.776 MPa, and 1.433 MPa were recorded, which were 10.19%, 10.27%, 10.20%, and 10.34% of the CS of LWAC. Besides, in the case of 56 curing days of the CS of LWAC. Besides, in the case of 56 curing days, the STS were recorded as

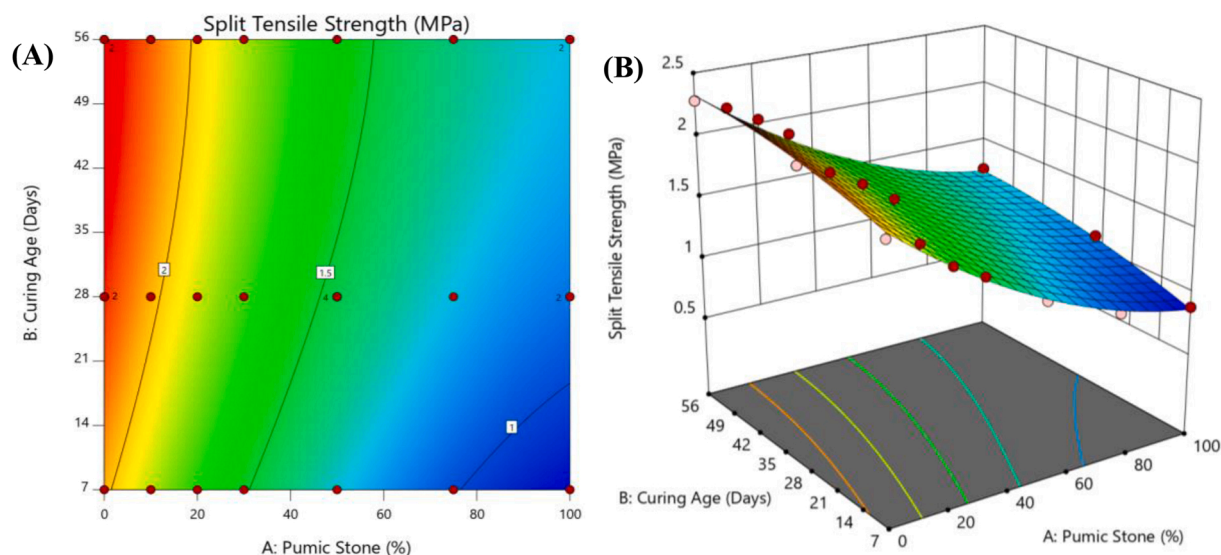


Fig. 15. Correlation between PS, curing age, and STS; (A) Contour graph; (B) 3D view.

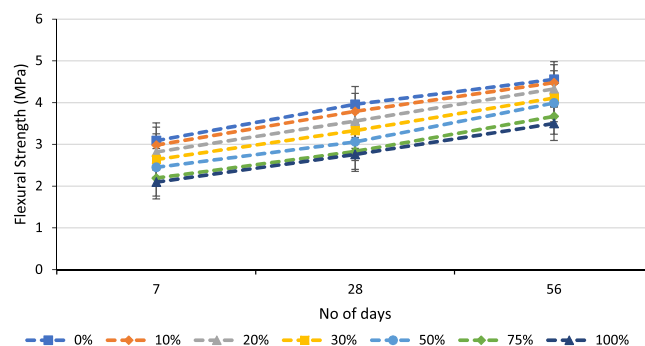


Fig. 16. Experimental results of FS containing PS at varying percentages and curing days.

2.18 MPa, 2.045 MPa, 1.88 MPa, and 1.537 MPa, which were 9.48%, 9.53%, 9.38%, and 9.33% of the CS of LWAC. Recent studies show that STS is 7–12% of the CS in conventional concrete [73,74]. Therefore, the current study concluded that the effect of lightweight aggregate concrete incorporating PS on STS is within the percentages (7–12%) of the CS for all the STS for 7 days, 28 days, and 56 days of curing with varying percentage replacement. Moreover, a linear relationship was found among the curing days, percentage replacement, and STS.

The statistical correlation between the CS and STS is shown in Fig. 13 in which Fig. 13 (a) and (b) represent the correlation for 7 days and 28 days, while (c) for 56 days. There is a strong correlation between the CS and STS for all the curing days with a coefficient of determination (R^2) values greater than 0.97 (97%) and having a linear second-degree polynomial expression.

Like CS, several graphs are generated through RSM for the STS to study the correlation between the variables, as shown in Fig. 14. The normal distribution of the data can be accessed from the points around the straight line. In the normal plot of distribution, as indicated in Fig. 14 (A), the points are close to the straight line validating that the data is normally distributed. However, compared to the depicted Residuals vs. Predicted as shown in (B) and (C) Predicted vs. Run, the Residuals vs. Actual are close to the straight line representing that the data generated from the runs are more likely to the data obtained from the laboratory experiments as revealed in Fig. 14 (D).

To graphically represent the influence of PS and curing age on the STS of lightweight aggregate concrete, the contour and 3D graphs are produced through RSM. Fig. 15 depicts the correlation between the curing days, PS, and STS in which (A) represents the contour graph, whereas (B) illustrates the 3D view of the lightweight aggregate concrete containing PS as a natural coarse aggregate substitute. The statistics show that increasing the percentage replacement of PS as a natural coarse aggregate replacement reduces the STS. However, the STS rises as the number of cure days grow from 7 to 56 days. The highest STS is attained on 56 days of drying with 0% and 10% replacement of PS. Moreover, with up to 30% of PS as a natural coarse aggregate, the concrete achieves the standard STS for all 7–56 days of drying.

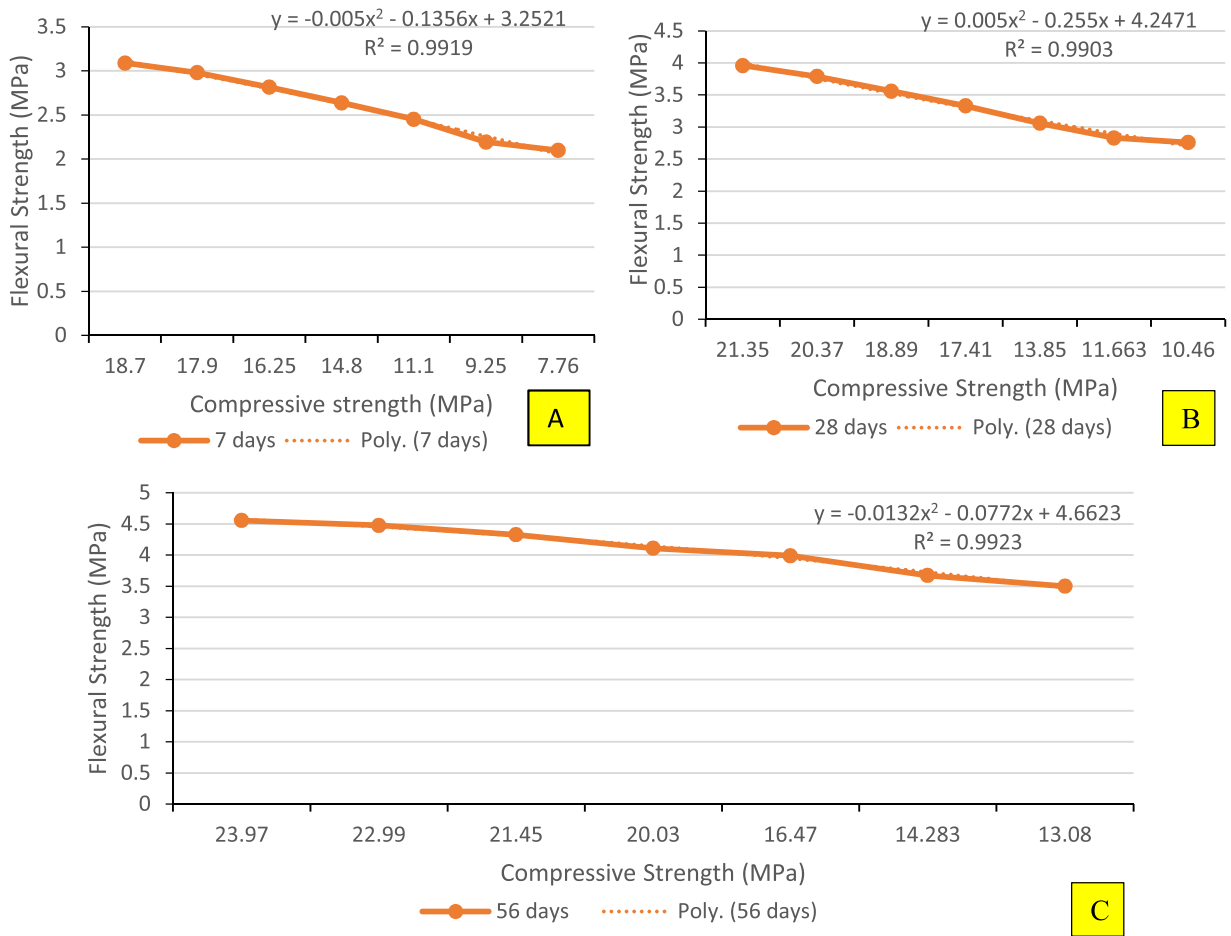


Fig. 17. Correlation between CS and FS a) 7-days b) 28-days c) 56-days.

3.5. Flexural strength (FS)

As per the ASTM C293, the FS of lightweight aggregate concrete including PS as a natural coarse aggregate substitute for 7–56 days of drying was investigated [63]. Compared to the concrete's STS, the FS of the concrete is essential. Fig. 16 depicts the experimental results of FS containing PS at varying percentages and curing days. Although the number of specimens and the curing days for the FS of concrete are similar to STS and CS, the beam's dimension varied. The prism beam was used to produce the samples for the FS with a dimension of $150 \times 150 \times 500$ mm.

As the proportion of the PS rises, the strength properties decrease; however, up to 30% substitute, the CS, STS, and FS for 7–56 days are within the ranges of LWA strength. The strength properties have a direct relationship with the curing days, increase in curing days can increase the strength properties. The highest was recorded on 56 days, which means that replacing the PS as natural coarse aggregate improves the FS but the opposite in CS. The FS on 7–56 days of drying with a 10%, 20%, 30%, 50%, 75%, and 100% replacement are within and greater than the standard limits for conventional concrete. Banawair et al., write a review on the strength properties of LWAC and show the relationship among FS and CS, where they concluded that the ratio of FS to CS is 11.6–13.5% whereas, the medium and high strength LWAC is typically within the range among 9–11% of the compressive strength [74].

In the current study, the ratio between CS and FS ranges from 16.65% to 25.03% for 7 days of curing and 18.63–26.38% on 28 days. However, for the 56 days of curing, it ranges from 19.46% to 26.76%. Therefore, this study concludes that the ratio of the CS to FS is increased by rising the proportion substitution of PS in the LWAC, whereas FS is the opposite. Lakshman et al., and Minapu et al., also achieved that the FS of concrete increases by increasing the replacement of natural aggregate with PS [75,76].

Fig. 17 depicts the statistical correlation among the FS and CS of lightweight aggregate concrete containing PS. The correlation of 7–56 days is indicated in Fig. 12 (a), (b), and (c). A strong correlation between the FS and CS of lightweight aggregate concrete including PS was found with an R^2 above 0.99 (99%) between the variables.

Same as CS and STS, the normality of the data is tested for the FS using RSM, and produced several graphs that show the normality of the data distribution for the FS as shown in Fig. 18. The normal plot of residuals, as presented in Fig. 18 (A), depicts that the points are in line with the straight line representing the high normality of the data. Besides, the predicted values produced from the design

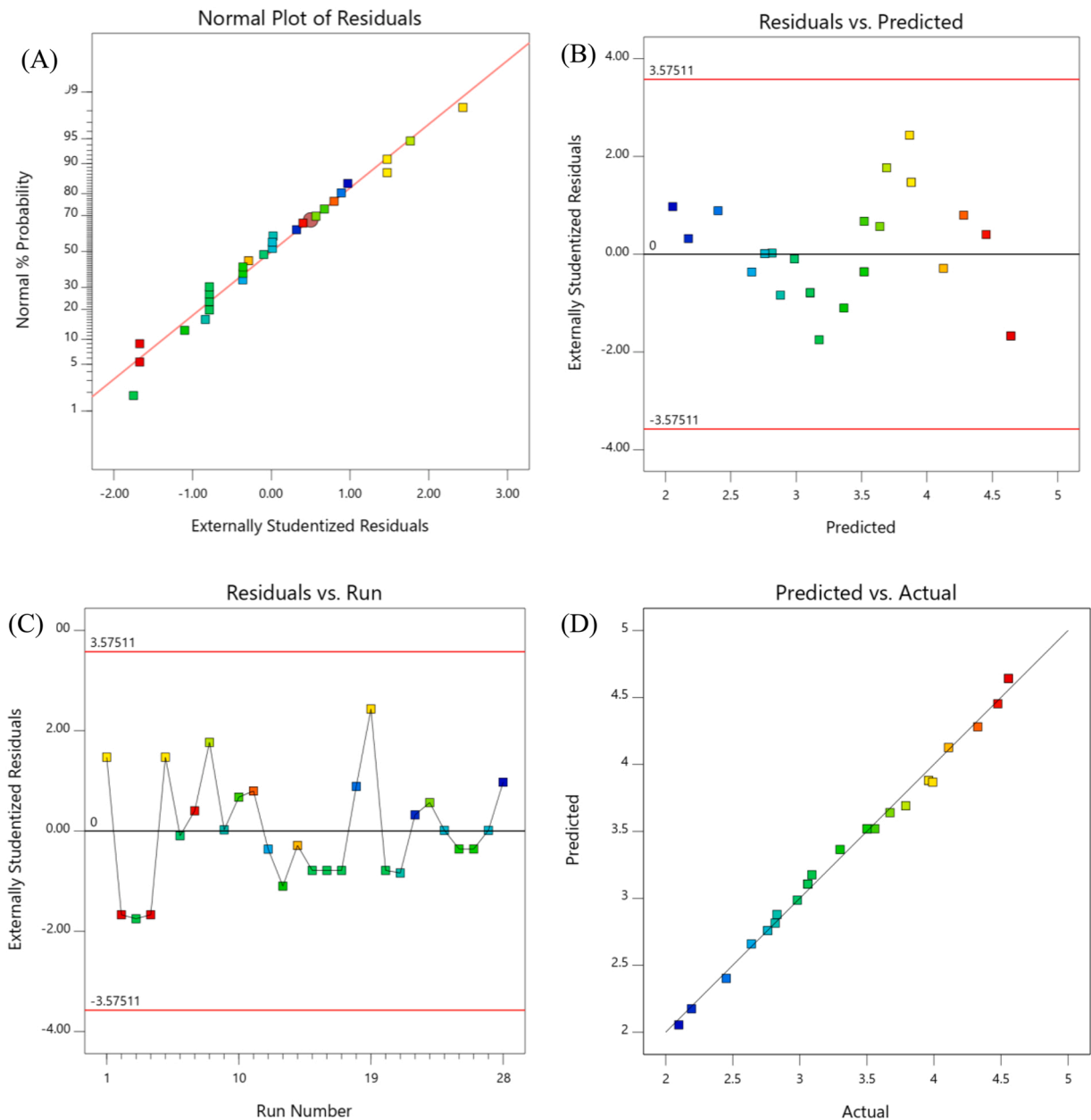


Fig. 18. Distribution of Data for STS; (A) Normal Plot of Residuals; (B) Predicted vs. Residuals; (C) Run vs. Predicted; (D) Actual vs. Residuals.

software and normally distributed with the actual values obtained from the laboratory experiments forming a straight line, as shown in Fig. 18 (D). Moreover, Fig. 18 (B) and (C) show the Predicted vs. Residuals and Run vs. Predicted for the FS of concrete incorporating PS as natural coarse aggregate replacement.

Both the contour and 3D graphs are produced for the FS of concrete using RSM to investigate the correlation between the input (curing ages and percentage replacement of PS) and output (FS) variables as shown in Fig. 19. The FS increased by raising the drying days from 7 to 56 days; however, it was the opposite with the PS. Up to 30% replacement of PS for all curing days, the FS of LWAC containing PS as a natural coarse aggregate substitute follows the standard limits of 9 – 11% of the CS. The ratio of the CS to FS is increased by rising the proportion substitute of PS in the LWAC for all curing days.

4. Model optimization and validation

Using the Design Expert 11.0 Program (Stat-Ease Inc., Minneapolis, USA), an optimization strategy was used to identify the best value of STS, FS, and CS for various ratios and curing days. During the optimization of the model, the targeted range (7 – 56 days of

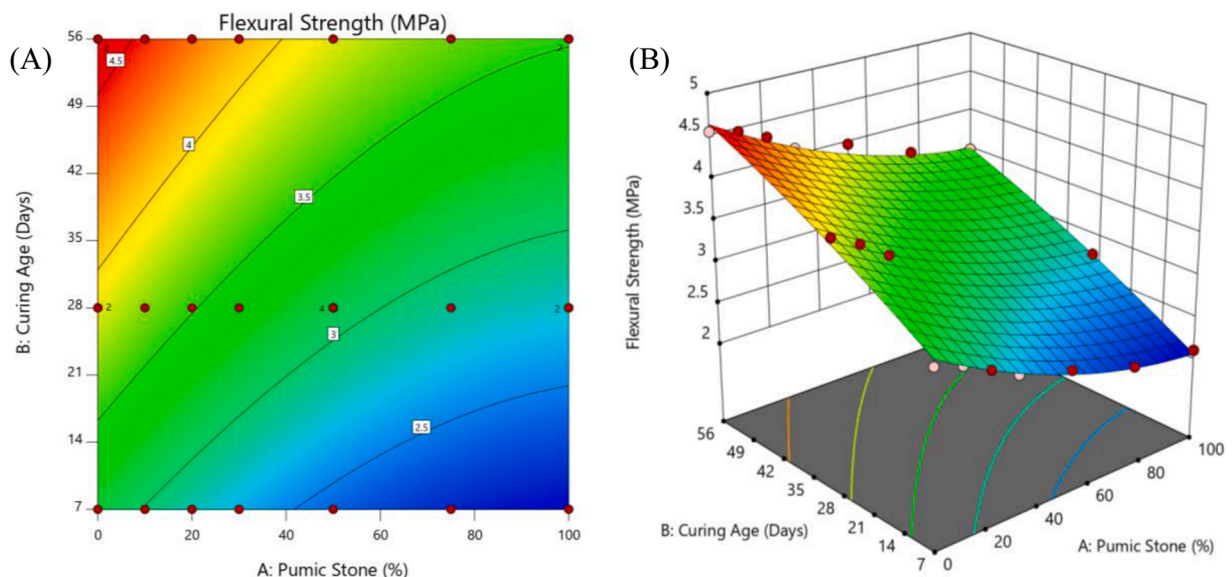


Fig. 19. Correlation between PS, curing age, and FS; (A) Contour graph; (B) 3D view.

Table 7

Optimum Combination for each age curing.

Pumice Stone (%)	Curing (Days)	CS (MPa)	STS (MPa)	FS (MPa)
16.505	7	16.42	1.73	2.873
23.064	28	17.8	1.819	3.47
29.335	56	19.576	1.84	4.136

curing) and ratio (0 – 100% replacement) were determined, as shown in Table 3 above (input parameters for the CCD). Table 7 depicts the optimum combination for each curing day and percentage replacement. The optimum PS for 7 days of curing is shown in Fig. 20 (A), which depicts that with 16.5% replacement on 7 days of curing, the CS of 16.42 MPa, STS of 1.73 MPa, and FS of 2.87 MPa can be achieved that are validated with the additional experiments.

On the other side, the optimum PS for 28 days of curing corresponding to CS, STS, and FS are shown in Fig. 20 (B), which depicts that with 23.064% replacement on 28 days of curing, the CS of 17.8 MPa, STS of 1.82 MPa, and FS of 3.47 MPa. Besides, as shown in Fig. 20 (C) and Table 7, the optimum PS is 29.335% for 56 curing days corresponding to CS, STS, and FS is 19.576 MPa for CS, 1.84 MPa for the STS, and 4.136 MPa for the FS.

The optimization and validation of the model results also show that up to 30% of the PS can be replaced for 7 – 56 curing days in which the CS is greater than 15 MPa, STS is 7–12% of the CS, and FS is 9 – 11% of the CS. Therefore, the optimum CS for the current study ranges from 17 to 20 MPa in which the STS is 9.39 – 10.54% of the CS and FS ranges from 17.49% to 21.12% of the CS.

5. Conclusion

The current study aims to replace the lightweight aggregate PS at a varying percentage as natural coarse aggregate to evaluate the strength properties of concrete incorporating PS at 7, 28, and 56 curing days. The statistical analysis and optimization of the RSM were used for the experimental design to design and execute the experiments based on the proposed run from RSM by adjusting the upper and lower boundaries of the drying days and PS percentage replacement as input parameters. The output parameters of the RSM were the strength properties such as STS, FS, and CS. After the laboratory experiments, the data is entered into RSM to generate a statistical equation, ANOVA, predicted vs. actual, contour, and 3D graphs for all the curing days and all the outputs, such as STS, FS, and CS. Below is a summary of the current study that is concluded as follows.

- There is a slight variation in the fresh properties of lightweight aggregate concrete incorporating PS. As the percentage of the PS increase, the slump values decreases. The slump cone test findings show that pumice lightweight aggregate concrete with 10% replacement has higher workability than the other replacement percentages. However, there is an increase and decrease in the water absorption properties of concrete containing PS due to the microstructure and internal properties of PS. In the current study, the highest water absorption was found on 50% replacement which is in line with the Kartika et al. studies.
- In terms of mechanical properties, the control mix with 0% replacement shows a high CS, STS, and FS than the rest of the percentages for all the curing days. However, this study concluded that concrete with up to 30% replacement of lightweight aggregate

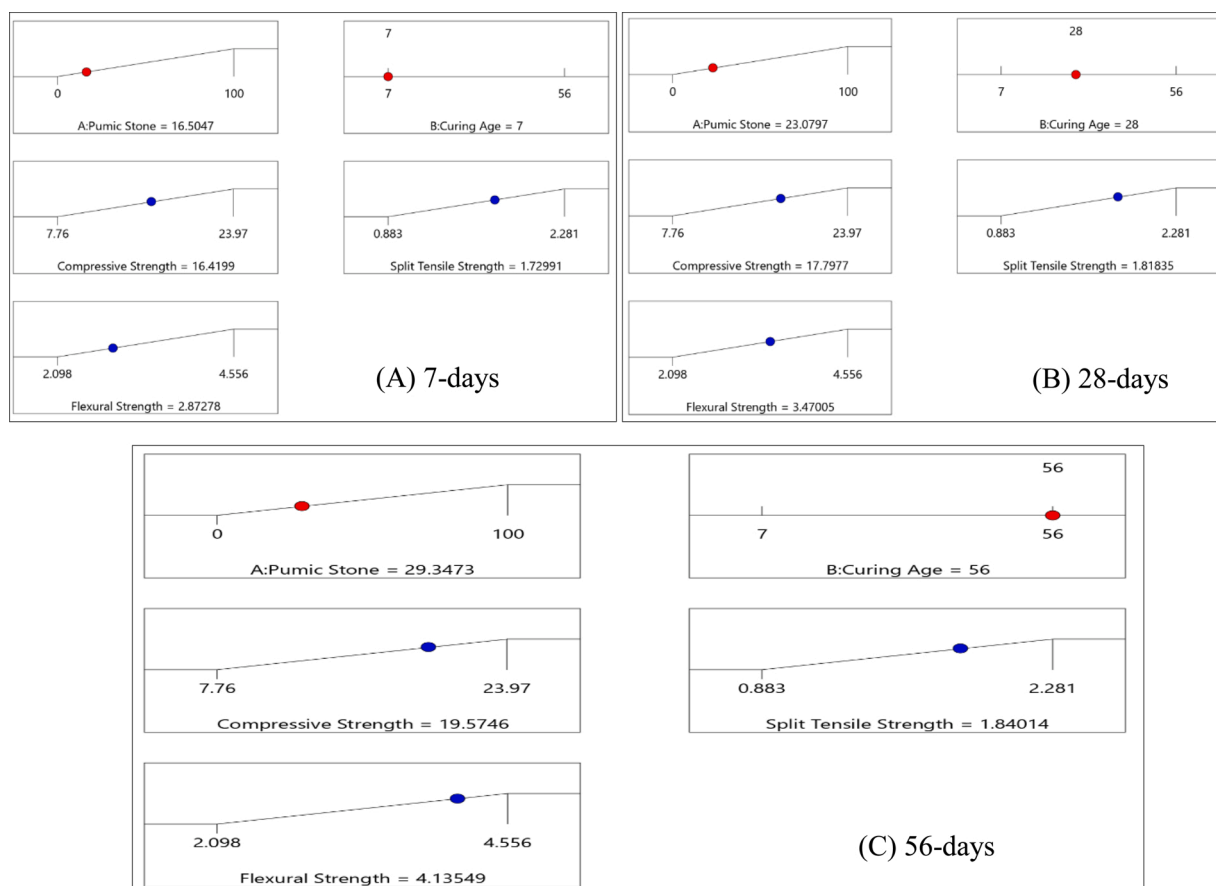


Fig. 20. The optimum percentage of PS for (A) 7-days (B) 28-days (C) 56-days.

PS could be used for structural purposes as the strength properties were higher than the standard limits of lightweight aggregate concrete, that is 15 MPa.

- According to the literature, the CS of lightweight aggregate concrete should be greater than 15 MPa in 7 days. However, in the current study, the CS on 7 days with 10%, 20%, and 30% replacement of PS were greater than 15 MPa and ranged from 17.9 to 14.9 MPa. Besides, the CS on 28 days of curing should be above 17 MPa for the conventional lightweight aggregate concrete, which is 20.37 – 17.41 MPa on 28 days and is found to be 22.99 – 20.33 MPa on 56 days of curing in the current study.
- In terms of STS, the STS is 8 – 12% of the CS, which in the current study ranges from 9% to 11% and was in the standard limits; however, it was found to be less than the control mix with 0% replacement of PS as a coarse aggregate replacement. Besides, the ratio of FS to CS is 9 – 11%; whereas the current study concluded 16.65 – 26.76%. Therefore, in the current study, the ratio between CS and FS ranges from 16.65% to 25.03% for 7 days of curing and 18.63–26.38% on 28 days. However, for the 56 days of curing, it ranges from 19.46% to 26.76%.
- The coefficient of variation (R^2) for all models was larger than 0.99 (99%), indicating the model has a high significance. Furthermore, an indicator of the model's relevance is that the difference among adjusted R^2 and R^2 is quite negligible, indicating that the model is highly significant.
- Finally, model validation was carried out by optimizing the model in RSM. The model validation revealed that up to 30% of the PS could be replaced for 7 – 56 curing days in which the CS is greater than 15 MPa, STS is 7–12% of the CS, and FS is 9 – 11% of the CS. Therefore, the optimum CS for the current study ranges from 17 to 20 MPa in which the STS is 9.39 – 10.54% of the CS, and FS ranges from 17.49% to 21.12% of the CS.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgments

The authors would like to thank King Saud University, Riyadh, Saudi Arabia, for researchers supporting project number RSP2023R145.

References

- [1] B.T.A. Manjunath, Partial replacement of E-plastic waste as coarse-aggregate in concrete, *Procedia Environ. Sci.* vol. 35 (2016) 731–739, <https://doi.org/10.1016/j.proenv.2016.07.079>.
- [2] J.K. Prusty, S.K. Patro, Properties of fresh and hardened concrete using agro-waste as partial replacement of coarse aggregate – a review, *Constr. Build. Mater.* vol. 82 (2015) 101–113, <https://doi.org/10.1016/j.conbuildmat.2015.02.063>.
- [3] M.J. Islam, M.S. Meherier, A.K.M.R. Islam, Effects of waste PET as coarse aggregate on the fresh and hardened properties of concrete, *Constr. Build. Mater.* vol. 125 (2016) 946–951, <https://doi.org/10.1016/j.conbuildmat.2016.08.128>.
- [4] B. Bahoria, D. Parbat, P. Naganaik, Replacement of natural sand in concrete by waste products: a state of art, *J. Environ. Res. Dev.* vol. 7 (4A) (2013) 1651.
- [5] A.U. Elinwa, Y.A. Mahmood, Ash from timber waste as cement replacement material, *Cem. Concr. Compos.* vol. 24 (2) (2002) 219–222, [https://doi.org/10.1016/S0958-9465\(01\)00039-7](https://doi.org/10.1016/S0958-9465(01)00039-7).
- [6] S. Chowdhury, M. Mishra, O. Suganya, The incorporation of wood waste ash as a partial cement replacement material for making structural grade concrete: an overview, *Ain Shams Eng. J.* vol. 6 (2) (2015) 429–437, <https://doi.org/10.1016/j.asej.2014.11.005>.
- [7] A.R.G. de Azevedo, M.T. Marvila, M. Ali, M.I. Khan, F. Masood, C.M.F. Vieira, Effect of the addition and processing of glass polishing waste on the durability of geopolymeric mortars, *Case Stud. Constr. Mater.* vol. 15 (2021), e00662, <https://doi.org/10.1016/j.cscm.2021.e00662>.
- [8] M. Usman, et al., Eco-friendly self-compacting cement pastes incorporating wood waste as cement replacement: a feasibility study, *J. Clean. Prod.* vol. 190 (2018) 679–688, <https://doi.org/10.1016/j.jclepro.2018.04.186>.
- [9] J. Torkaman, A. Ashori, A. Sadr Momtazi, Using wood fiber waste, rice husk ash, and limestone powder waste as cement replacement materials for lightweight concrete blocks, *Constr. Build. Mater.* vol. 50 (2014) 432–436, <https://doi.org/10.1016/j.conbuildmat.2013.09.044>.
- [10] S. Hesami, A. Modarres, M. Soltaninejad, H. Madani, Mechanical properties of roller compacted concrete pavement containing coal waste and limestone powder as partial replacements of cement, *Constr. Build. Mater.* vol. 111 (2016) 625–636, <https://doi.org/10.1016/j.conbuildmat.2016.02.116>.
- [11] A.A. Alani, et al., Demolition waste potential for completely cement-free binders, *Materials* vol. 15 (17) (2022) 6018 [Online]. Available, (<https://www.mdpi.com/1996-1944/15/17/6018>).
- [12] E.D. Barreto, et al., Clay ceramic waste as pozzolan constituent in cement for structural concrete, *Materials* vol. 14 (11) (2021), <https://doi.org/10.3390/ma14112917>.
- [13] M. Samadi, et al., Properties of mortar containing ceramic powder waste as cement replacement, *J. Teknol.* vol. 77 (12) (2015).
- [14] B. Bhardwaj, P. Kumar, Waste foundry sand in concrete: a review, *Constr. Build. Mater.* vol. 156 (2017) 661–674, <https://doi.org/10.1016/j.conbuildmat.2017.09.010>.
- [15] B. Taha, G. Nounu, Utilizing waste recycled glass as sand/cement replacement in concrete, *J. Mater. Civ. Eng.* vol. 21 (12) (2009) 709–721, [https://doi.org/10.1061/\(ASCE\)0899-1561\(2009\)21:12\(709\)](https://doi.org/10.1061/(ASCE)0899-1561(2009)21:12(709)).
- [16] M. Imran Khan, et al., Investigating the mechanical properties and fuel spillage resistance of semi-flexible pavement surfacing containing irradiated waste PET based grouts, *Constr. Build. Mater.* vol. 304 (2021), 124641, <https://doi.org/10.1016/j.conbuildmat.2021.124641>.
- [17] Z.C. Steyn, A.J. Babafemi, H. Fataar, R. Combrinck, Concrete containing waste recycled glass, plastic and rubber as sand replacement, *Constr. Build. Mater.* vol. 269 (2021), 121242, <https://doi.org/10.1016/j.conbuildmat.2020.121242>.
- [18] F.S. Khalid, N.B. Azmi, K.A.S.M. Sumandi, P.N. Mazenan, Mechanical properties of concrete containing recycled concrete aggregate (RCA) and ceramic waste as coarse aggregate replacement, in: *AIP Conference Proceedings*, vol. 1891, AIP Publishing LLC, 2017.
- [19] T.S. Thandavamoorthy, Wood waste as coarse aggregate in the production of concrete, *Eur. J. Environ. Civ. Eng.* vol. 20 (2) (2016) 125–141, <https://doi.org/10.1080/19648189.2015.1016631>.
- [20] M. Saltan, F.Selcan Findik, Stabilization of subbase layer materials with waste pumice in flexible pavement, *Build. Environ.* vol. 43 (4) (2008) 415–421, <https://doi.org/10.1016/j.buildenv.2007.01.007>.
- [21] M.A. Sultan, A. Gaus, R. Hakim, Review of the flexural strength of lightweight concrete beam using pumice stone as of substitution partial coarse aggregate, *GEOMATE J.* vol. 21 (85) (2021) 154–159.
- [22] J.J. Shideler, Lightweight-aggregate concrete for structural use, *J. Proc.* vol. 54 (10) (1957) 299–328.
- [23] G.C. Hoff, Fire resistance of high-strength concretes for offshore concrete platforms, *Spec. Publ.* vol. 163 (1996) 53–88.
- [24] A. Nafees, et al., Modeling of mechanical properties of silica fume-based green concrete using machine learning techniques, *Polymers* vol. 14 (1) (2022) 30.
- [25] T. Gjerde, Structural lightweight-aggregate concrete (LWA-concrete) for marine and offshore applications, *Nor. Contract.* (1982).
- [26] S. Room, M. Ali, M.A. Alam, U. Khan, S. Ammad, S. Saad, Assessment of lightweight aggregate concrete using textile washing stone, 2021 Third Int. Sustain. Resil. Conf.: Clim. Change (2021) 327–333, <https://doi.org/10.1109/IEEECONF53624.2021.9668076>.
- [27] M. Ali, F. Masood, M.I. Khan, M. Azeem, F.N. Ali, Evaluation of flexible pavement distresses - a case study of Northern Bypass Peshawar, Pakistan, 15–16 Nov. 2021, 2021 Third Int. Sustain. Resil. Conf.: Clim. Change (2021) 399–404, <https://doi.org/10.1109/IEEECONF53624.2021.9668173>.
- [28] L. XIAOPENG, "Structural lightweight concrete with pumice aggregate," 2005.
- [29] K.M.A. Hossain, Properties of volcanic pumice based cement and lightweight concrete, *Cem. Concr. Res.* vol. 34 (2) (2004) 283–291.
- [30] M. Ali, S. Abbas, A.R.G. de Azevedo, M.T. Marvila, M.I. Khan, W. Rafiq, Experimental and analytical investigation on the confinement behavior of low strength concrete under axial compression, *Structures* vol. 36 (2022) 303–313, <https://doi.org/10.1016/j.istruc.2021.12.038>.
- [31] M.A. Idi, A.S. Abdulazeez, S. Usman, T. Justin, Strength properties of concrete using pumice aggregate as partial replacement of coarse aggregate, *Int. J. Eng. Appl. Sci. Technol.* vol. 4 (11) (2020) 519–525.
- [32] M. Ali, et al., Investigating optimal confinement behaviour of low-strength concrete through quantitative and analytical approaches, *Materials* vol. 14 (16) (2021), <https://doi.org/10.3390/ma14164675>.
- [33] M. Ali, S. Abbas, M.I. Khan, M.A. Gad, S. Ammad, A. Khan, "Experimental Validation of Mander's Model for Low Strength Confined Concrete Under Axial Compression," in *2020 Second International Sustainability and Resilience Conference: Technology and Innovation in Building Designs(51154)*, Sakheer, Bahrain, 11–12 Nov. 2020 2020, pp. 1–6, doi: <https://doi.org/10.1109/IEEECONF51154.2020.9319950>.
- [34] V. Lesovik, et al., Four-component high-strength polymineral binders, *Constr. Build. Mater.* vol. 316 (2022), 125934, <https://doi.org/10.1016/j.conbuildmat.2021.125934>.
- [35] V. Lesovik, A. Tolstoy, R. Fediuk, M. Amran, M. Ali, A.R.G. de Azevedo, Improving the performances of a mortar for 3D printing by mineral modifiers, *Buildings* vol. 12 (8) (2022) 1181 [Online]. Available, (<https://www.mdpi.com/2075-5309/12/8/1181>).
- [36] K. Liu, et al., Optimization of autogenous shrinkage and microstructure for ultra-high performance concrete (UHPC) based on appropriate application of porous pumice, *Constr. Build. Mater.* vol. 214 (2019) 369–381, <https://doi.org/10.1016/j.conbuildmat.2019.04.089>.

- [37] R.B. Karthika, V. Vidyapriya, K.V. Nandhini Sri, K. Merlin Grace Beaula, R. Harini, M. Sriram, Experimental study on lightweight concrete using pumice aggregate, /01/01/ 2021, *Mater. Today: Proc.* vol. 43 (2021) 1606–1613, <https://doi.org/10.1016/j.matpr.2020.09.762>.
- [38] M.H. Zhang, O.E. Gjovrv, Mechanical properties of high-strength lightweight concrete, *Mater. J.* vol. 88 (3) (1991) 240–247.
- [39] T. Parhizkar, M. Najimi, A.R. Pourkhorshidi, Application of pumice aggregate in structural lightweight concrete 2012.
- [40] B.D. Pravalika, K.V. Rao, The study on strength properties of light weight concrete using light weight aggregate, *Int. J. Sci. Res.* vol. 5 (6) (2016) 1735–1739.
- [41] G. Venkateswararao, P.R. Kishore, A. Kumar, M. Yadav, Experimental Study on Strength Properties of Light Weight High Strength Fibre Reinforced Concrete With Partial Replacement of Coarse Aggregate With Pumice Stone.
- [42] M. Wang, X. Yang, W. Wang, Establishing a 3D aggregates database from X-ray CT scans of bulk concrete, *Constr. Build. Mater.* vol. 315 (2022), 125740, <https://doi.org/10.1016/j.conbuildmat.2021.125740>.
- [43] Z. Dong, W. Qian, X. Ma, X. Li, J. Zhou, Asymptotic homogenization of effective thermal-elastic properties of concrete considering its three-dimensional mesostructure, *Comput. Struct.* vol. 279 (2023), 106970, <https://doi.org/10.1016/j.compstruc.2022.106970>.
- [44] H. Huang, M. Li, Y. Yuan, H. Bai, Experimental research on the seismic performance of precast concrete frame with replaceable artificial controllable plastic hinges, *J. Struct. Eng.* vol. 149 (1) (2023), 04022222, <https://doi.org/10.1061/JSENDH.STENG-11648>.
- [45] H. Huang, Y. Yao, C. Liang, Y. Ye, Experimental study on cyclic performance of steel-hollow core partially encased composite spliced frame beam, /12/01/ 2022, *Soil Dyn. Earthq. Eng.* vol. 163 (2022), 107499, <https://doi.org/10.1016/j.soildyn.2022.107499>.
- [46] Y. Huang, W. Zhang, X. Liu, Assessment of diagonal macrocrack-induced debonding mechanisms in FRP-strengthened RC beams, *J. Compos. Constr.* vol. 26 (5) (2022), 04022056, [https://doi.org/10.1061/\(ASCE\)CC.1943-5614.0001255](https://doi.org/10.1061/(ASCE)CC.1943-5614.0001255).
- [47] C. Zhang, H. Khorshidi, E. Najafi, M. Ghasemi, Fresh, mechanical and microstructural properties of alkali-activated composites incorporating nanomaterials: a comprehensive review, *J. Clean. Prod.* vol. 384 (2023), 135390, <https://doi.org/10.1016/j.jclepro.2022.135390>.
- [48] W. Zhang, S. Kang, Y. Huang, X. Liu, Behavior of reinforced concrete beams without stirrups and strengthened with basalt fiber-reinforced polymer sheets, /04/ 01 2023, *J. Compos. Constr.* vol. 27 (2) (2023), 04023007, <https://doi.org/10.1061/JCOF2.CCENG-4082>.
- [49] M. Ali, et al., Central composite design application in the optimization of the effect of waste foundry sand on concrete properties using RSM, /12/01/ 2022, *Structures* vol. 46 (2022) 1581–1594, <https://doi.org/10.1016/j.istruc.2022.11.013>.
- [50] M.I. Khan, et al., Cementitious grouts for semi-flexible pavement surfaces - a review, *Materials* vol. 15 (15) (2022) 5466 ([Online]. Available), (<https://www.mdpi.com/1996-1944/15/15/5466>).
- [51] M. Al Salaaheen, et al., Modelling and optimization for mortar compressive strength incorporating heat-treated fly oil shale ash as an effective supplementary cementitious material using response surface methodology, *Materials* vol. 15 (19) (2022) 6538.
- [52] R. Fediuk, M. Ali, Recyclable materials for ecofriendly technology, *Materials* vol. 15 (20) (2022) 7133 ([Online]. Available), (<https://www.mdpi.com/1996-1944/15/20/7133>).
- [53] A.M. Memon, et al., Physicochemical, rheological and morphological properties of bitumen incorporating petroleum sludge, /08/23/ 2021, *Constr. Build. Mater.* vol. 297 (2021), 123738, <https://doi.org/10.1016/j.conbuildmat.2021.123738>.
- [54] *ASTM Standard Specification for Portland Cement*, ASTM-C150/C150M, American Society for Testing and Materials, 2020.
- [55] Standard Test Methods for Time of Setting of Hydraulic Cement by Vicat Needle., ASTM-C191–19, 2019.
- [56] Standard Test Method for Relative Density (Specific Gravity) and Absorption of Fine Aggregate, ASTM-C128–15, American Society for Testing and Materials, 2015.
- [57] Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates., ASTM-C136/C136M-19, 2019.
- [58] A.M. Rashad, An overview of pumice stone as a cementitious material – the best manual for civil engineer, /02/01 2021, *Silicon* vol. 13 (2) (2021) 551–572, <https://doi.org/10.1007/s12633-020-00469-3>.
- [59] A.I.M. Ismail, O.I. El-Shafey, M.H.A. Amr, M.S. El-Maghraby, Pumice characteristics and their utilization on the synthesis of mesoporous minerals and on the removal of heavy metals, 259379-259379, (Eng.), *Int Sch. Res Not.* vol. 2014 (2014), <https://doi.org/10.1155/2014/259379>.
- [60] M. Malakootian, M. Moosazadeh, N. Yousefi, A. Fatehizadeh, Fluoride removal from aqueous solution by pumice: case study on Kuhbonan water, *Afr. J. Environ. Sci. Technol.* vol. 5 (4) (2011) 299–306.
- [61] Standard test method for the compressive strength of concrete cylinder, ASTM-C39/C39M, 2020.
- [62] Standard Test Method for Splitting Tensile Strength of Cylindrical Concrete Specimens, A. C496, West Conshohocken, PA, USA, 2017.
- [63] Standard Test Method for Flexural Strength of Concrete (Using Simple Beam With Center-Point Loading), A. C293/C293M—16, American Society for Testing and Materials: West Conshohocken, PA, USA, 2015.
- [64] L. Gündüz, The effects of pumice aggregate/cement ratios on the low-strength concrete properties, /05/01/ 2008, *Constr. Build. Mater.* vol. 22 (5) (2008) 721–728, <https://doi.org/10.1016/j.conbuildmat.2007.01.030>.
- [65] M.O. KALE, H. ÇAĞLAR, A. ÇAĞLAR, A.C. APAY, S. ÇİMEN, Improving of lightweight concrete properties produced with pumice aggregate of nevsehir region with fly ash substitution, *Acad. Platf. -J. Eng. Sci.* vol. 9 (2) (2021) 302–308.
- [66] P. Lehner, P. Konečný, P. Ghosh, Variation of durability and strength parameters of pumice based mixtures, *Materials* vol. 14 (13) (2021) 3674 ([Online]. Available), (<https://www.mdpi.com/1996-1944/14/13/3674>).
- [67] T. Wu, X. Yang, H. Wei, X. Liu, Mechanical properties and microstructure of lightweight aggregate concrete with and without fibers, /02/28/ 2019, *Constr. Build. Mater.* vol. 199 (2019) 526–539, <https://doi.org/10.1016/j.conbuildmat.2018.12.037>.
- [68] R. Committee, "Functional Classification of Lightweight Concrete. Recommendation LC 2 11," ed: Rilem Paris, 1978.
- [69] S. Hedjazi, "Compressive strength of lightweight concrete," *Compressive Strength of Concrete*, pp. 1–8.
- [70] M. Ali, D.B.E. Dharmawijoyo, A.R.G. de Azevedo, R. Fediuk, H. Ahmad, B. Salah, Time-use and spatio-temporal variables influence on physical activity intensity, physical and social health of travelers, *Sustainability* vol. 13 (21) (2021) 12226, <https://doi.org/10.3390/su132112226>.
- [71] M. Ali, et al., The influence of COVID-19-induced daily activities on health parameters—a case study in Malaysia, *Sustainability* vol. 13 (2021) 7465, <https://doi.org/10.3390/su13137465>.
- [72] M. Ali, et al., Assessment of local earthen bricks in perspective of physical and mechanical properties using geographical information system in Peshawar, Pakistan, December, 1, *Structures* vol. 28 (2020) 2549–2561, <https://doi.org/10.1016/j.istruc.2020.10.075>.
- [73] D.T.C. Lee, T.S. Lee, The effect of aggregate condition during mixing on the mechanical properties of oil palm shell (OPS) concrete, *MATEC Web Conf.* vol. 87 (2017) 01019 (EDP Sciences).
- [74] A. Singh, A. Kumar, S. Singh, H. Singh, Study of partial replacement of fine aggregate by iron slag, *Int. J. Adv. Res.* vol. 4 (7) (2016) 687–702.
- [75] A. Lakshman, H.B. Devi, M. Dominic, Influence of lightweight aggregates on various properties of concrete, *Int. J. Eng. Res. Tech.* vol. 8 (2) (2019) 82–85.
- [76] L.K. Minapu, M. Ratnam, U. Rangaraju, Experimental study on light weight aggregate concrete with pumice stone, silica fume and fly ash as a partial replacement of coarse aggregate, *Int. J. Innov. Res. Sci., Eng. Technol.* vol. 3 (12) (2014) 18130–18138.